

CAP5735 Data Visualization and Reproducible Research:

Data Visualization Mini Project 2 - NBAchampionsdata.csv

a.What were the original charts you planned to create for these assignments? What steps were necessary for cleaning and preparing the data?

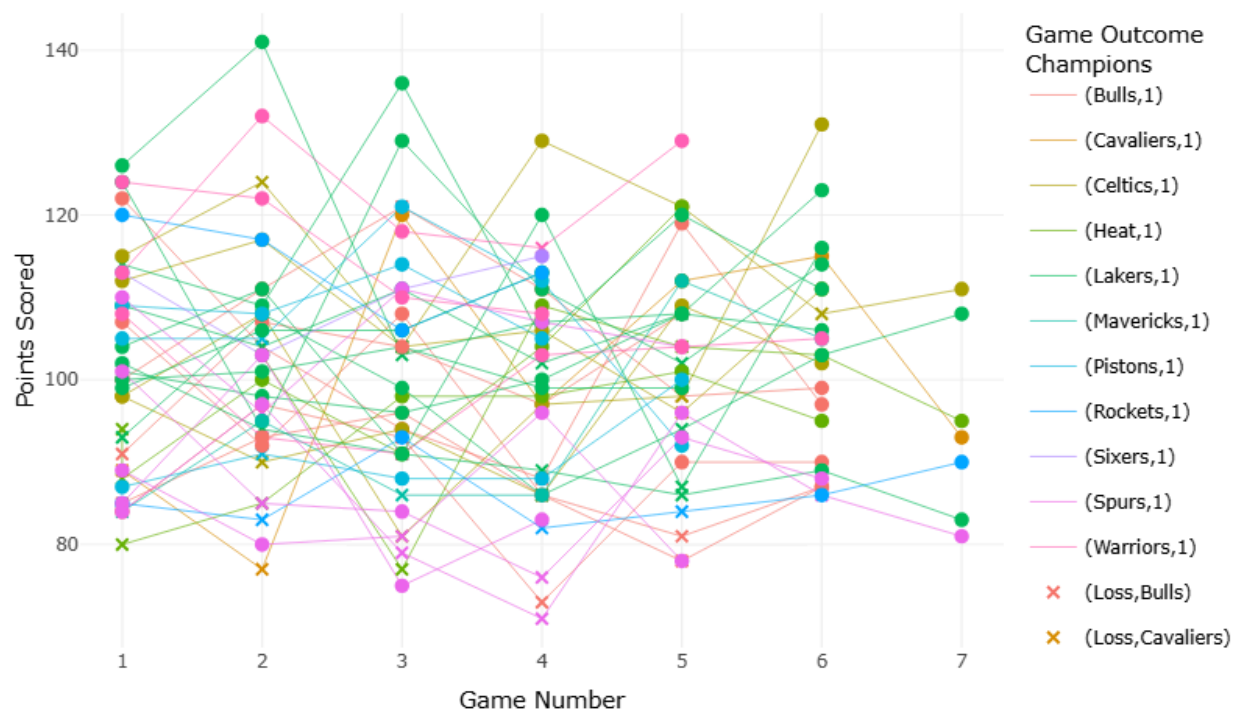
I planned on mapping the games with an interactive chart as it would allow for exploration of each team individually rather than be a jumble of every team at once. For the map I planned on creating a map of where each team is from, allowing for easy understanding for people who may not recognize teams just based on name. The last graph I wanted to do was a linear regression based on stats to see what stats are most impactful on points scored.

The dataset had 2 entries with the team name input wrong one had 'Heat' (with the single quotes) and another had Warriorr instead of Warriors, I manually fixed these before doing my analysis.

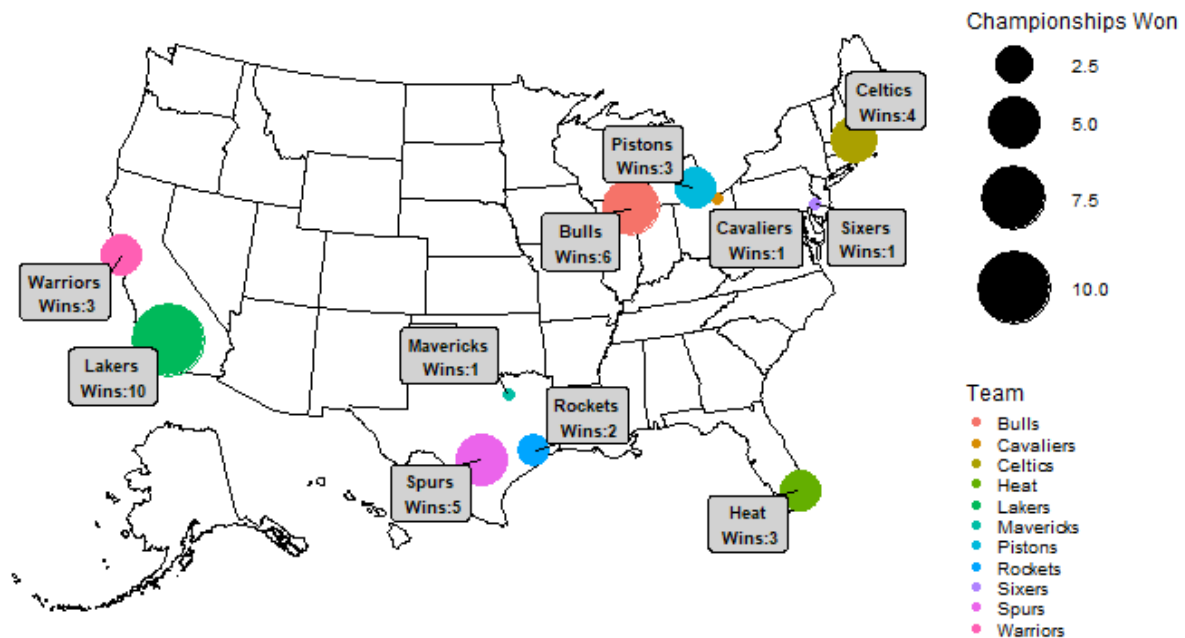
The map also didn't have city location data so I utilized Google Gemini to find all of the city coordinates, saving me from doing each one manually.

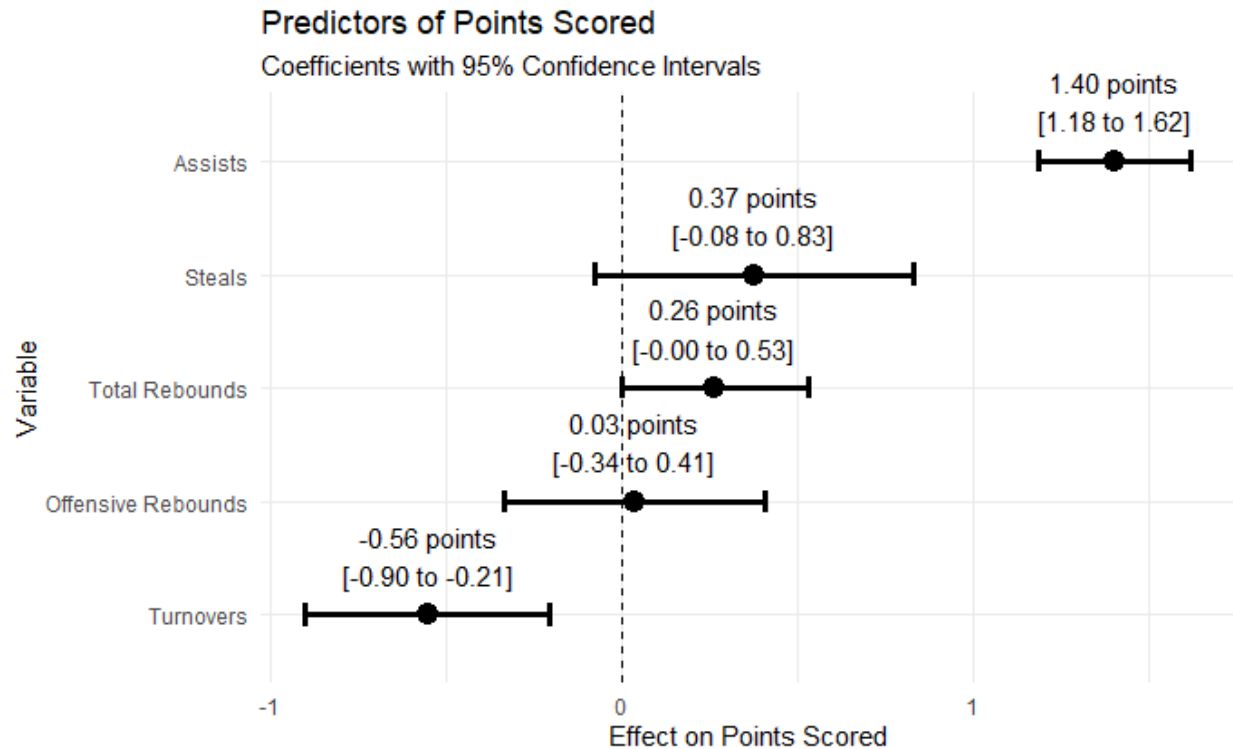
b.What story could you tell with your plots? What difficulties did you encounter while creating the visualizations? What additional approaches do you think can be used to explore the data you selected?

NBA Finals Games (1980–2018)



NBA Championships by Team Location (1980–2018)





The interactive plot was very interesting to me as it showed the correlation between scoring and winning each game. Most of the lost games are near the bottom which are the lower point games. However, since only the team that won the overall finals are shown almost every loss is followed by a much higher scoring win, indicating that these championship teams try even harder following a loss and usually succeed.

The visualizations showed some interesting results, looking at the map shows how dominant regions are with Texas and California winning 21 of the 38 championships despite there being dozens of other teams. This surprised me even though I knew that teams like the Lakers won a lot in this timeframe. This could be due to the high population of those areas, which means more fans and more income allowing them to pay better players more to join their team over another.

The linear regression also yielded interesting results, assists were the most impactful on point scoring which makes sense because an assist can only happen if someone scores. More interestingly however, offensive rebounds had a smaller impact than total rebounds. I would have thought the opposite as an offensive rebound is almost always another scoring opportunity, whereas a defensive one wouldn't necessarily turn into a scoring chance.

When annotating the map I had a very hard time trying to get the labels to not overlap, `geom_sf_label` wouldn't allow me to prevent overlap without manually setting each label

which I didn't want to do and also didn't feel like a good solution if I ever needed to make changes. After some research I landed on the `ggrepel` package using `geom_label_repel`, as I could keep most of the label settings that I had already set with just some small changes rather than having to fully restart. I also had the issue of the legend getting cut off on the bottom or top which I fixed by using `legend.key.size`.

I think that tracking stats across the years, such as three pointers, steals, fouls, etc. could be used to show how basketball has changed from shooting more threes to more/less physical or aggressive playstyles. This would be interesting for this dataset specifically because I have a feeling, for example, that once a team that shot more 3's won the championships more teams would follow and more winners would be shooting more 3's.

c.How did you apply the principles of data visualizations and design for this assignment?

I ensured that the colors of the teams were consistent between the interactive graph and the map, ensuring that there wouldn't be confusion when moving from one graph to another.

When annotating I made sure to keep the annotations small, readable, and only display key information. Along with keeping labels from overlapping, this kept the graphs readable and allowed for the easy understanding of the important takeaways.

Utilizing the `theme_minimal` kept the charts and graphs free from clutter while still keeping the important data readable, helping to focus attention on the data.